

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0720

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks: 50

Annexure A

Scenario-Based

10% 43

Code of Conduct:

(192511488)

I Varsha Vaedhini (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S. Vaedhini

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → /24

Quality Assurance → /25

Human Resources → /26

Executive Management → /27

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the /64 prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.

3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context(2)	Shows good understanding with minor gaps(1.5)	Partial understanding; misses key aspects(1)	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved(2)	Identifies most key issues with minor omissions(1.5)	Identifies limited issues; some irrelevant points(1)	Unable to identify key issues
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Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale(2)	Makes reasonable decisions with some justification(1.5)	Makes weak decisions with limited justification(1)	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented(1.5)	Generally clear with minor issues(1)	Somewhat unclear or poorly structured(0.5)	Disorganized and difficult to understand

	Understanding of Scenario (20%)	Identification of key issue (20%)	Application of concepts (25%)	Decision making (20%)	clarity of Response (15%)	
Q1	2	1.5	1.5	1.5	1.5	8
Q2	1.5	1	2	1	1	6.5
Q3	1	2	2.5	1	1	7.5
Q4	1	2	1	1.5	1.5	7
Q5	1.5	1	1.5	1	1	6

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Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

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1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

ANSWER :

Given

- Network: **192.168.100.0/24**
- Computing: **/25**
- Electronics: **/26**
- Mechanical Engineering: **/27**

Formulae Used

1. Number of host bits : $\text{host bits} = 32 - \text{prefix length}$
2. Total number of addresses : $\text{total address} = 2^{\text{host bits}} - 2$
3. Usable host addresses : $2^{\text{host bits}} - 2$
4. Usable hosts = 256 – last octet of subnet mask

School of Computing — /25

Subnet mask : /25 = 255.255.255.128

Step 1: Calculate host bits

$$\text{Host bits} = 32 - 25 = 7 \text{ bits}$$

Step 2: Calculate total IP addresses

$$\text{Total IP addresses} = 2^7 = 128 \text{ addresses}$$

Step 3: Calculate usable host addresses

$$\text{Usable hosts} = 128 - 2 = 126 \text{ hosts}$$

Step 4: Calculate block size

$$\text{Block size} = 256 - 128 = 128$$

Therefore, the first subnet ranges from : 192.168.100.0 to 192.168.100.127

Therefore:

Network Address = 192.168.100.0

Valid Host Range = 192.168.100.1 to 192.168.100.126

Broadcast Address = 192.168.100.127

Usable Host Addresses = 126

School of Electronics – /26

The previous subnet ends at 192.168.100.127.

Therefore, the next available address is 192.168.100.128.

Subnet mask for /26 = 255.255.255.192

Host bits = $32 - 26 = 6$ bits

Total IP addresses = $2^6 = 64$ addresses

Usable hosts = $64 - 2 = 62$ hosts

Block size = $256 - 192 = 64$

Starting from 192.168.100.128, the subnet ranges from : 192.168.100.128 to
192.168.100.191

Therefore:

Network Address = 192.168.100.128

Valid Host Range = 192.168.100.129 to 192.168.100.190

Broadcast Address = 192.168.100.191

Usable Host Addresses = 62

School of Mechanical Engineering – /27

The previous subnet ends at 192.168.100.191.

Therefore, the next available address is 192.168.100.192.

Subnet mask for /27 = 255.255.255.224

Host bits = $32 - 27 = 5$ bits

Total IP addresses = $2^5 = 32$ addresses

Usable hosts = $32 - 2 = 30$ hosts

Block size = $256 - 224 = 32$

Starting from 192.168.100.192, the subnet ranges from : 192.168.100.192 to
192.168.100.223

Therefore:

Network Address = 192.168.100.192

Valid Host Range = 192.168.100.193 to 192.168.100.222

Broadcast Address = 192.168.100.223

Usable Host Addresses = 30

Remaining Unused IP Address Range

The original /24 network ends at : 192.168.100.255

The Mechanical Engineering subnet ends at : 192.168.100.223

Therefore, the next available address is : 192.168.100.224

Hence, the remaining unused IP range is : 192.168.100.224 to 192.168.100.255

Department	Subnet	Network Address	Valid Host IP Range	Broadcast Address	Usable Host Addresses
School of Computing	/25	192.168.100.0	192.168.100.1 – 192.168.100.126	192.168.100.127	126
School of Electronics	/26	192.168.100.128	192.168.100.129 – 192.168.100.190	192.168.100.191	62
School of Mechanical Engineering	/27	192.168.100.192	192.168.100.193 – 192.168.100.222	192.168.100.223	30

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**

Quality Assurance → **/25**

Human Resources → **/26**

Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

ANSWERS :

Given

- Software Development → /24
- Quality Assurance → /25
- Human Resources → /26
- Executive Management → /27

Formulae Used

- Host bits = $32 - \text{Prefix length}$
- Total addresses = $2^{(\text{Host bits})}$
- Usable hosts = Total addresses - 2
- Block size = 256 - Last octet of subnet mask

Software Development – /24

$$\text{Subnet mask} = 255.255.255.0$$

$$\text{Host bits} = 32 - 24 = 8$$

$$\text{Total addresses} = 2^8 = 256$$

$$\text{Usable hosts} = 256 - 2 = 254$$

$$\text{Block size} = 256 - 0 = 256$$

Therefore:

- Network Address = 10.10.0.0
- Valid Host Range = 10.10.0.1 – 10.10.0.254
- Broadcast Address = 10.10.0.255
- Usable Hosts = 254

Quality Assurance – /25

The previous subnet ends at 10.10.0.255, so the next available address is
10.10.1.0.

Subnet mask = 255.255.255.128

Host bits = $32 - 25 = 7$

Total addresses = $2^7 = 128$

Usable hosts = $128 - 2 = 126$

Block size = $256 - 128 = 128$

Therefore:

- Network Address = 10.10.1.0
- Valid Host Range = 10.10.1.1 – 10.10.1.126
- Broadcast Address = 10.10.1.127
- Usable Hosts = 126

Human Resources – /26

The previous subnet ends at 10.10.1.127, so the next available address is
10.10.1.128.

Subnet mask = 255.255.255.192

Host bits = $32 - 26 = 6$

Total addresses = $2^6 = 64$

Usable hosts = $64 - 2 = 62$

Block size = $256 - 192 = 64$

Therefore:

- Network Address = 10.10.1.128
- Valid Host Range = 10.10.1.129 – 10.10.1.190
- Broadcast Address = 10.10.1.191
- Usable Hosts = 62

Executive Management – /27

The previous subnet ends at 10.10.1.191, so the next available address is
10.10.1.192.

Subnet mask = 255.255.255.224

$$\text{Host bits} = 32 - 27 = 5$$

$$\text{Total addresses} = 2^5 = 32$$

$$\text{Usable hosts} = 32 - 2 = 30$$

$$\text{Block size} = 256 - 224 = 32$$

Therefore:

- Network Address = 10.10.1.192
- Valid Host Range = 10.10.1.193 – 10.10.1.222
- Broadcast Address = 10.10.1.223
- Usable Hosts = 30

Department	Subnet	Network Address	Valid Host IP Range	Broadcast Address	Usable Host Addresses
Software Development	/24	10.10.0.0	10.10.0.1 – 10.10.0.254	10.10.0.255	254
Quality Assurance	/25	10.10.1.0	10.10.1.1 – 10.10.1.126	10.10.1.127	126
Human Resources	/26	10.10.1.128	10.10.1.129 – 10.10.1.190	10.10.1.191	62
Executive Management	/27	10.10.1.192	10.10.1.193 – 10.10.1.222	10.10.1.223	30

Original IP Block	Allocated Up To	Remaining Unused IP Range
10.10.0.0/16	10.10.1.223	10.10.1.224 – 10.10.255.255

3.A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the /64 prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

ANSWER :

Given IPv6 network: **2001:0db8:abcd:0000:0000:0000:0000/64**

Compressed notation

- Remove leading zeros in each group and replace the consecutive groups of all zeros with ::.
- Answer: 2001:db8:abcd::/64

Expanded notation

- The compressed address : 2001:db8:abcd::/64
- contains three written groups followed by ::.
- :: represents five groups of 0000 here.
- Therefore : 2001:0db8:abcd:0000:0000:0000:0000/64

Network Prefix

- A /64 means the first 64 bits are the network portion.

- The first four groups contain 64 bits : 2001:0db8:abcd:0000
- Therefore: Network Prefix = 2001:0db8:abcd:0000/64
- Host portion = remaining 64 bits

Total Host Addresses

- IPv6 has 128 bits.
- Network bits = 64
- Host bits: $128 - 64 = 64$ bits
- Formula: Total host addresses = $2^{(\text{host bits})}$
- $= 2^{64} = 18,446,744,073,709,551,616$

Total Possible Host Addresses

Since the question asks for the total possible host addresses within this /64 network, it is:

$$2^{64} = 18,446,744,073,709,551,616 \text{ addresses}$$

Unlike IPv4, we do not subtract 2 for network and broadcast addresses in the usual IPv6 host-count calculation.

Question	Answer
1. Compressed notation	2001:db8:abcd::/64
2. Expanded notation	2001:0db8:abcd:0000:0000:0000:0000:0000/64
3. Network prefix	2001:0db8:abcd:0000/64
4. Host bits	$128 - 64 = 64$ bits
5. Total host addresses	$2^{64} = 18,446,744,073,709,551,616$

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

ANSWER :

The router has these routing entries:

- 192.168.1.0/24
- 192.168.2.0/24
- 192.168.3.0/24

Destination IP : 192.168.2.45

Destination Subnet

- The destination IP is : 192.168.2.45
- With /24, the first three octets represent the network : 192.168.2.0

- Therefore : Destination subnet = 192.168.2.0/24

Matching Routing Table Entry

Compare the destination IP with the routing table:

192.168.1.0/24 → No match
192.168.2.0/24 → MATCH ✓
192.168.3.0/24 → No match

Therefore : Matching entry = 192.168.2.0/24

Next-Hop Network / Interface

The packet belongs to the 192.168.2.0/24 network.

Therefore, the router forwards the packet through the interface connected to the 192.168.2.0/24 customer network.

Next-hop network/interface = 192.168.2.0/24 interface

The question does not provide a specific next-hop IP address or interface name, so we should not invent one.

Default Route

If the destination does not match any of the listed routing entries, the router uses a default route.

The standard representation is : 0.0.0.0/0

It means : Match any IPv4 destination that has no more specific route in the routing table.

Requirement	Answer
1. Destination Subnet	192.168.2.0/24
2. Matching Routing Table Entry	192.168.2.0/24
3. Next-Hop Network/Interface	Interface connected to 192.168.2.0/24
4. Default Route	0.0.0.0/0

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

ANSWER :

Given

- Administration LAN = 192.168.10.0/26
- Finance LAN = 192.168.10.64/26
- Administration gateway = 192.168.10.1
- Finance Gateway = 192.168.10.65

Formulae Used

- Host bits = 32 – Prefix length
- Total addresses = $2^{(\text{Host bits})}$
- Usable host addresses = Total addresses – 2
- Block size = 256 – Last octet of subnet mask

For /26:

Subnet mask = 255.255.255.192

Host bits : $32 - 26 = 6$ bits

Total addresses: $2^6 = 64$ addresses

Usable host addresses: $64 - 2 = 62$ hosts

Block size: $256 - 192 = 64$

Therefore, each /26 subnet contains 64 addresses, with 62 usable host addresses.

Administration LAN – 192.168.10.0/26

The block size is 64, so the first subnet is: 192.168.10.0 – 192.168.10.63

Therefore:

- Network Address: 192.168.10.0
- Valid Host Range: 192.168.10.1 – 192.168.10.62
- Broadcast Address: 192.168.10.63
- Default Gateway: 192.168.10.1

Three Suitable Computers

Since .1 is already used as the gateway, suitable host addresses are:

- Computer 1 → 192.168.10.2
- Computer 2 → 192.168.10.3
- Computer 3 → 192.168.10.4

Finance LAN – 192.168.10.64/26

The next /26 block starts at .64: 192.168.10.64 – 192.168.10.127

Therefore:

- Network Address: 192.168.10.64
- Valid Host Range: 192.168.10.65 – 192.168.10.126
- Broadcast Address: 192.168.10.127
- Default Gateway: 192.168.10.65

Three Suitable Computers

Since .65 is already used as the gateway:

- Computer 1 → 192.168.10.66
- Computer 2 → 192.168.10.67
- Computer 3 → 192.168.10.68

LAN	Network Address	Valid Host IP Range	Broadcast Address	Default Gateway
Administration	192.168.10.0	192.168.10.1 – 192.168.10.62	192.168.10.63	192.168.10.1
Finance	192.168.10.64	192.168.10.65 – 192.168.10.126	192.168.10.127	192.168.10.65

LAN	Computer 1	Computer 2	Computer 3
Administration	192.168.10.2	192.168.10.3	192.168.10.4
Finance	192.168.10.66	192.168.10.67	192.168.10.68

Usable hosts per LAN: 62

Administration Gateway: 192.168.10.1

Finance Gateway: 192.168.10.65

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